Comprehensive Methods for Exponential Decay Analysis, Probability Thresholding, and Confidence Intervals in Feeding Behavior

January 23, 2025

1 Introduction

This document outlines the detailed methods used to analyze pellet retrieval data in mice, focusing on:

- **Clustering** of feeding events based on Inter-Pellet Intervals (IPI).
- **Exponential decay** modeling of cluster-size frequency.
- **Conditional probabilities** of cluster growth.
- **Confidence interval** estimation for these probabilities.
- **Threshold determination** using a Maximum Gap Method.

We illustrate how these steps were used to define biologically relevant categories such as "snacks," "meals," and "mega-meals," culminating in a refined understanding of mouse feeding behavior.

2 Data Acquisition and Preprocessing

2.1 Timestamp Recording and Conversion

Mice were housed with `FED3` devices that dispensed and logged pellets. Each pellet retrieval time was recorded in *hours* from the start of the experiment:

$$p_i^{(\text{hours})}, \quad i = 1, 2, \dots, N.$$

For uniformity and to facilitate time calculations, we converted all timestamps to seconds:

$$p_i^{(\text{seconds})} = p_i^{(\text{hours})} \times 3600.$$

We then sorted all $p_i^{(\text{seconds})}$ in ascending order.

2.2 Selection Criteria for Mice

A total of 11 mice were included in the final dataset, based on specific criteria (e.g., sex, order, dietary phase). Only these 11 mice's data were analyzed in the subsequent steps.

3 Inter-Pellet Intervals and Clustering

3.1 Definition of IPI

After sorting timestamps p_i, we defined the Inter-Pellet Interval (IPI) between consecutive pellets i and $i+1$:

$$\mathrm{IPI}_i \;=\; p_{i+1} - p_i, \quad \text{for } i = 1, 2, \ldots, (N-1).$$

This provided a measure of the time gap between one pellet retrieval and the next.

3.2 Clustering Threshold

To decide whether two consecutive pellets belong to the same "feeding event," we selected a threshold T (e.g., $T = 60$ seconds):

$$\text{if } \mathrm{IPI}_i \leq T \implies \text{Pellet } (i+1) \text{ remains in the same cluster as Pellet } i,$$

$$\text{if } \mathrm{IPI}_i > T \implies \text{Start a new cluster at Pellet } (i+1).$$

Iterating this logic partitions the full sequence of pellets into discrete clusters (feeding events):

$$\{\text{Cluster}_1,\ \text{Cluster}_2,\ \ldots\}.$$

The *size* of each cluster is the number of pellets grouped within it.

4 Cluster Size Distribution

For each mouse, we computed how many times the mouse exhibited a cluster of size 1, size 2, size 3, and so forth. Summing (or averaging) across mice yields the overall **cluster size distribution**. Let $n_i(x)$ be the number of events of size x for mouse i. If we have N mice:

$$y(x) \;=\; \frac{1}{N} \sum_{i=1}^{N} n_i(x),$$

which is the mean frequency of cluster size x across all selected mice.

5 Exponential Decay Model

5.1 Why Exponential Decay?

Empirically, larger feeding bouts (i.e., many pellets in one event) are typically less frequent than smaller bouts. An exponential decay function naturally captures this decreasing frequency:

$$y(x) \;=\; a\, e^{-\,b\,x},$$

where $y(x)$ is the average count of events of size x, a is the scale parameter, and b is the decay rate parameter.

5.2 Fitting Procedure

We used nonlinear least squares to fit (a, b) such that the residual sum of squares

$$\mathrm{SSR}(a,b) = \sum_x \Big(y(x) \; - \; a\,e^{-b\,x}\Big)^2$$

is minimized. Here, $y(x)$ is the *observed* mean frequency of cluster size x, and $a\,e^{-b\,x}$ is the model prediction.

5.3 Goodness of Fit (R^2)

We define:

$$R^2 = 1 \; - \; \frac{\sum_x \big[y(x) - a\,e^{-bx}\big]^2}{\sum_x \big[y(x) - \overline{y}\big]^2},$$

where $\overline{y}$ is the average of the observed $y(x)$ values. An R^2 near 1 indicates a strong fit. In this study:

$$a = 139.55, \quad b = 0.28, \quad R^2 = 0.9668.$$

That is, approximately 96.68% of the variation in $y(x)$ is explained by the exponential decay.

5.4 Interpreting a and b

- $\boldsymbol{a = 139.55}$ sets the overall scale of the distribution. Extrapolated to $x = 0$, $y(0) \approx a$.
- $\boldsymbol{b = 0.28}$ indicates how quickly the frequency decays with cluster size x. Each additional pellet multiplies the expected frequency by $e^{-b} \approx e^{-0.28} \approx 0.76$, implying a $\sim 24\%$ decrease per extra pellet.

6 Probability Interpretation and Conditional Growth

6.1 Probabilities of Cluster Sizes

While $y(x)$ represents *counts*, they can be normalized to probabilities by dividing by the total counts. If x ranges up to some maximum $X_{\max}$,

$$P(x) \;=\; \frac{y(x)}{\sum_{j=1}^{X_{\max}} y(j)}.$$

Thus, $P(x)$ is the probability of encountering a cluster of size x among all feeding events.

6.2 Conditional Probability $P(X + 1 \mid X)$

Another question is: *If a cluster has size X, what is the probability it will reach size $X + 1$?* Empirically, we define:

$$P(X + 1 \mid X) \;=\; \frac{n_{X+1}}{n_X},$$

where n_X is the total number of clusters of size X, and n_{X+1} is the total number of clusters of size $X + 1$. This yields a measure of how likely a feeding bout is to "grow" by one additional pellet.

7 Confidence Intervals for Conditional Probabilities

7.1 Wilson Interval (Example)

To assess the statistical certainty around $P(X+1 \mid X)$, we can calculate confidence intervals using, for instance, the Wilson formula. Suppose we observe $k = n_{X+1}$ "successes" out of $n = n_X$ "trials." The *Wilson interval* for a confidence level $1 - \alpha$ is:

$$\hat{p} = \frac{k}{n}, \quad \mathrm{CI}_{\mathrm{low/high}} = \frac{1}{1 + z_\alpha^2/n}\left[\hat{p} + \frac{z_\alpha^2}{2n} \;\pm\; z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z_\alpha^2}{4n^2}}\right],$$

where z_α is the critical value from the normal distribution (e.g., $z_{0.025} \approx 1.96$ for 95% CI). Hence, each $P(X + 1|X)$ is accompanied by a *lower* and *upper* confidence bound.

8 Threshold Determination and the Maximum Gap Method

8.1 Motivation

Sometimes we wish to find a threshold in $P(X+1|X)$ that differentiates between "likely to continue growing" vs. "unlikely to continue." This threshold can help define boundaries for *snacks* vs. *meals*, or *meals* vs. *mega-meals*.

8.2 Maximum Gap Method

Let $P(x + 1|x)$ be computed for $x = 1$ through some maximum (e.g., $x = 9$ if we only consider up to size 10). Sort these probabilities in ascending or cluster-size order, and define consecutive differences:

$$\Delta_i = \big[P(x+1 \mid x)\big]_{\,i+1} - \big[P(x+1 \mid x)\big]_{\,i}.$$

The *largest negative gap* often indicates a natural break in probability values. That is, a big drop in $P(X + 1|X)$ suggests a transition from a high-likelihood region to a low-likelihood region. We set the threshold just above or below that largest gap.

9 Refining Meal Definitions: Snacks, Meals, and Mega-Meals

9.1 Applying the Threshold

With a threshold P_{th}, we can classify feeding clusters by size. For example:

- **Snacks:** 1-pellet events (or those with $P(X+1|X)$ well below P_{th} starting at $X = 1$).
- **Meals:** Intermediate sizes (e.g., 2–10 pellets), as long as $P(X+1|X)$ remains above P_{th} up to a point, then drops below it.
- **Mega-Meals:** Clusters above some high-size cutoff (e.g., > 10 pellets), or persisting growth probabilities.

10 Summary of Results

10.1 Exponential Decay Fit

In this dataset of 11 mice, the exponential model

$$y(x) \;=\; 139.55\, e^{-\,0.28\,x}$$

achieved $R^2 = 0.9668$. This indicates that as x (number of pellets) increases, the expected frequency $y(x)$ decreases by roughly $e^{-0.28} \approx 0.76$ for each additional pellet.

10.2 Conditional Probability Insights

Calculating $P(X+1|X)$ showed that clusters have a decreasing likelihood of gaining another pellet with increasing size. Confidence intervals (via Wilson or similar methods) confirm statistical reliability of these conditional probabilities, and a *maximum gap* analysis can identify clear probability thresholds for delineating different feeding event categories.

10.3 Behavioral Relevance

These methods reveal that:

- Mice predominantly consume small feeding bouts, with larger bouts decreasing in frequency exponentially.
- A clear threshold in $P(X+1|X)$ can distinguish truly large meals (*mega-meals*) from typical smaller meals, highlighting points of behavioral transition.

11 Conclusion and Future Directions

By integrating:

- **IPI-based Clustering** (defining feeding events),
- **Exponential Decay Fitting** (capturing how often events of varying sizes occur),
- **Conditional Probability and Confidence Intervals** (quantifying growth likelihood from X to $X + 1$ pellets),
- **Maximum Gap Thresholding** (classifying events into snacks, meals, or mega-meals),

we achieve a comprehensive framework for analyzing rodent feeding behavior. Future work can explore differences in these parameters (decay rate, probability thresholds) across genotypes, diets, or times of day to further elucidate the biological and behavioral underpinnings of feeding patterns.